

# A. Linear Equations

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## Outcomes

- *Graph linear equations.*
- *Find the slope of a line.*
- *Determine equations of lines.*
- *Solve linear systems.*
- *Apply linear equations to real-world problems.*

## Chapter Overview

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In this chapter<sup>1</sup>, we focus on linear equations, their graphical representations, finding slopes, determining line equations from given information, and solving systems of linear equations. We also examine applications of these concepts in real-world scenarios, such as cost modeling, demand-supply equilibrium, and break-even analysis.

## A.1 Graphing a Linear Equation

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## Outcomes

- *Graph a line from its equation.*
- *Graph a line given its equation in parametric form.*
- *Graph and find equations of vertical and horizontal lines.*

Equations whose graphs are straight lines are called *linear equations*. Examples include:

$$2x - 3y = 6, \quad 3x = 4y - 7, \quad y = 2x - 5, \quad 2y = 3, \quad x - 2 = 0.$$

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<sup>1</sup>This section was modified and remixed from Applied Finite Mathematics (Sekhon and Bloom)/Section 1.1: Graphing a linear equation which was shared under a CC BY 4.0 license at [https://math.libretexts.org/Bookshelves/Applied\\_Mathematics/Applied\\_Finite\\_Mathematics\\_\(Sekhon\\_and\\_Bloom\)/01%3A\\_Linear\\_Equations/1.01%3A\\_Graphing\\_a\\_Linear\\_Equation](https://math.libretexts.org/Bookshelves/Applied_Mathematics/Applied_Finite_Mathematics_(Sekhon_and_Bloom)/01%3A_Linear_Equations/1.01%3A_Graphing_a_Linear_Equation)

A line is determined by two points. To graph a linear equation, we often choose values for one variable and solve for the other.

**Example A.1: Graphing a Line**

*Graph the line:*  $y = 3x + 2$ .

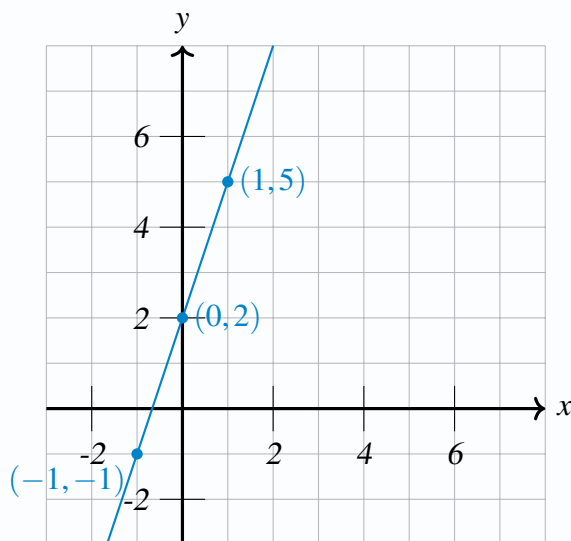
**Solution:** Choose convenient  $x$ -values:

$$x = -1 \implies y = 3(-1) + 2 = -1 \implies (-1, -1)$$

$$x = 0 \implies y = 3(0) + 2 = 2 \implies (0, 2)$$

$$x = 1 \implies y = 3(1) + 2 = 5 \implies (1, 5)$$

*Plot these points and draw the line.*

**Example A.2: Graphing Another Line**

*Graph the line:*  $2x + y = 4$ .

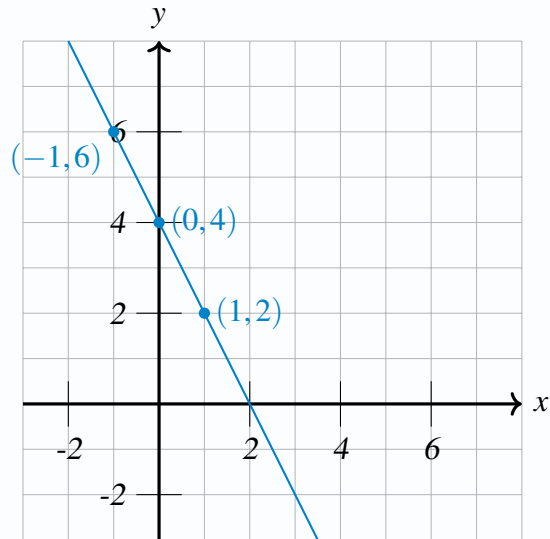
**Solution:** Choose points:

$$x = -1 \implies 2(-1) + y = 4 \implies y = 6 \implies (-1, 6)$$

$$x = 0 \implies y = 4 \implies (0, 4)$$

$$y = 2 \implies 2x + 2 = 4 \implies x = 1 \implies (1, 2)$$

*Plotting these points gives:*



## Intercepts

The  $x$ -intercept occurs where  $y = 0$ , and the  $y$ -intercept occurs where  $x = 0$ .

### Example A.3: Finding Intercepts

Find the intercepts of the line  $2x - 3y = 6$  and graph it.

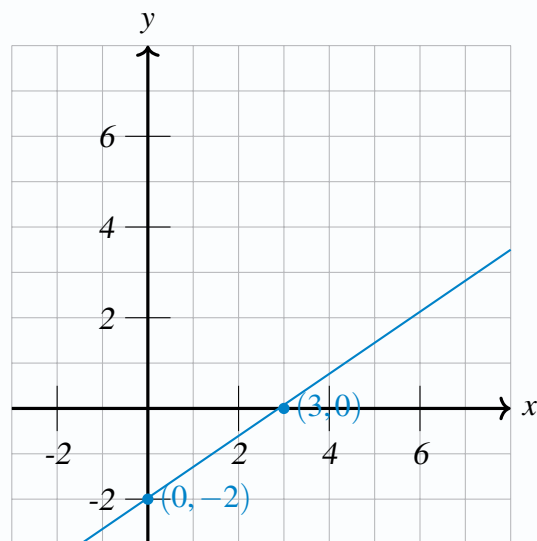
**Solution:** For the  $x$ -intercept, set  $y = 0$ :

$$2x - 3(0) = 6 \implies x = 3 \implies (3, 0).$$

For the  $y$ -intercept, set  $x = 0$ :

$$2(0) - 3y = 6 \implies y = -2 \implies (0, -2).$$

Plot these intercepts and draw the line.



## Parametric Form

Lines can also be given in parametric form, e.g.  $x = 3 + 2t$ ,  $y = 1 + t$ .

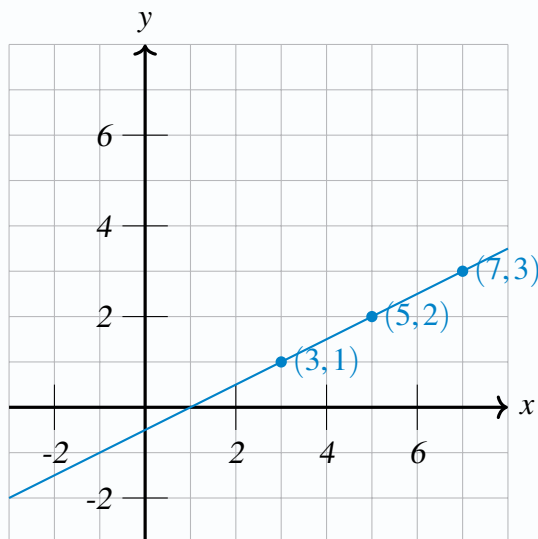
### Example A.4: Parametric Equations

Graph the line  $x = 3 + 2t$ ,  $y = 1 + t$ .

**Solution:** Let  $t = 0, 1, 2$ :

$$t = 0 \implies (x, y) = (3, 1), \quad t = 1 \implies (5, 2), \quad t = 2 \implies (7, 3).$$

Plot these points and draw the line.



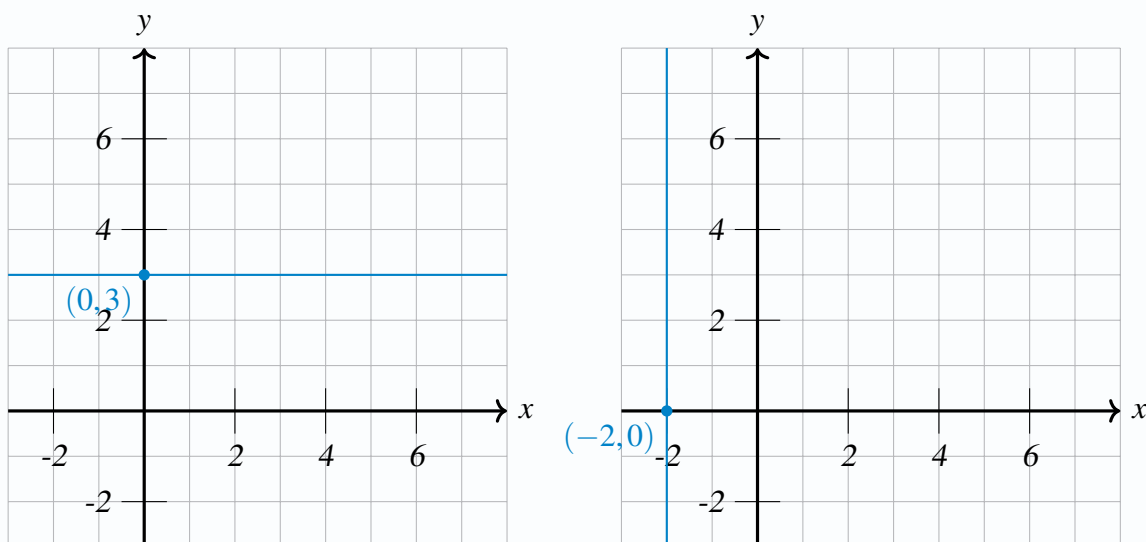
## Horizontal and Vertical Lines

$x = a$  is a vertical line passing through  $(a, 0)$ ,  $y = b$  is a horizontal line passing through  $(0, b)$ .

### Example A.5: Horizontal and Vertical Lines

Graph  $x = -2$  and  $y = 3$ .

**Solution:** The line  $x = -2$  is vertical through  $(-2, 0)$ . The line  $y = 3$  is horizontal through  $(0, 3)$ .



## A.2 Slope of a Line

### Outcomes

- Find the slope of a line given two points.
- Graph a line if a point and slope are given.

The slope  $m$  of a line passing through  $(x_1, y_1)$  and  $(x_2, y_2)$  is:

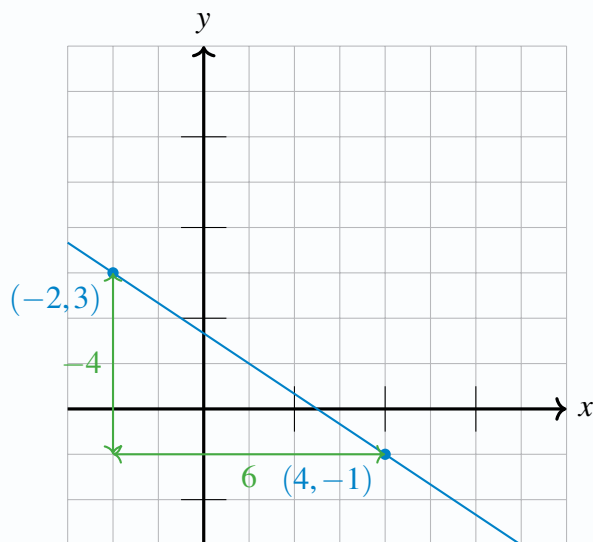
$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

### Example A.6: Finding Slope

Find the slope of the line through  $(-2, 3)$  and  $(4, -1)$ .

**Solution:**

$$m = \frac{-1 - 3}{4 - (-2)} = \frac{-4}{6} = -\frac{2}{3}.$$



Vertical lines have undefined slope; horizontal lines have slope 0.

### Example A.7: Vertical and Horizontal Slopes

a) For points  $(2, 3)$  and  $(2, -1)$ :

$$m = \frac{-1 - 3}{2 - 2} = \frac{-4}{0} \text{ (undefined), vertical line.}$$

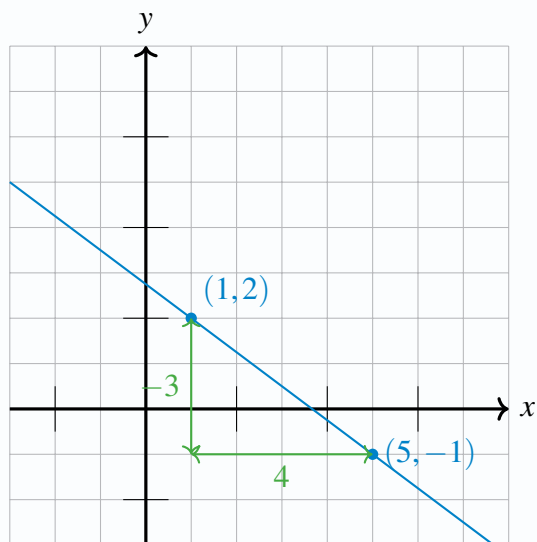
b) For points  $(-1, -4)$  and  $(3, -4)$ :

$$m = \frac{-4 - (-4)}{3 - (-1)} = \frac{0}{4} = 0 \text{ (horizontal line).}$$

### Example A.8: Graphing a Line from a Point and Slope

Graph the line passing through  $(1, 2)$  with slope  $-\frac{3}{4}$ .

**Solution:** Starting from  $(1, 2)$ , a slope of  $-\frac{3}{4}$  means down 3, right 4 to get another point  $(5, -1)$ . Plot and draw the line.



If a line is written as  $y = mx + b$ , the coefficient  $m$  is the slope and  $b$  is the  $y$ -intercept.

## A.3 Determining the Equation of a Line

### Outcomes

- Find an equation of a line given a point and slope.
- Find an equation of a line given two points.

### Forms of a Line

Slope-Intercept:  $y = mx + b$ , Point-Slope:  $y - y_1 = m(x - x_1)$ , Standard:  $Ax + By = C$ .

#### Example A.9: Equation Given Slope and Intercept

Find the equation of a line with slope  $m = 5$  and  $y$ -intercept  $b = 3$ .

**Solution:**  $y = 5x + 3$ .

#### Example A.10: Equation Given a Point and Slope

Find the equation of the line passing through  $(2, 7)$  with slope 3.

**Solution:**

$$y = 3x + b, 7 = 3(2) + b, b = 1 \implies y = 3x + 1.$$

**Example A.11: Equation Given Two Points**

Find the equation of the line passing through  $(-1, 2)$  and  $(1, 8)$ .

**Solution:**

$$m = \frac{8 - 2}{1 - (-1)} = \frac{6}{2} = 3, y = 3x + b.$$

Use one point:

$$2 = 3(-1) + b \implies b = 5, \text{ so } y = 3x + 5.$$

**Example A.12: Using Intercepts**

Find the equation of the line with  $x$ -intercept 3 and  $y$ -intercept 4.

**Solution:** Intercepts give points  $(3, 0)$  and  $(0, 4)$ :

$$m = \frac{4 - 0}{0 - 3} = -\frac{4}{3}, y = -\frac{4}{3}x + 4.$$

## A.4 Applications of Linear Equations

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**Outcomes**

- Use linear functions to model real-world applications.

Linear equations commonly model cost, revenue, and population changes.

**Example A.13: Cost Function**

A taxi charges \$0.50 per mile plus a \$5 flat fee.

**Solution:** Let  $x$  = miles,  $y$  = cost:

$$y = 0.50x + 5.$$

At  $x = 20$ :  $y = 0.50(20) + 5 = 15$ .

**Example A.14: Cost from Two Points**

It costs \$750 to make 25 items and \$1000 to make 50 items. Assume linearity.

**Solution:** Points  $(25, 750)$  and  $(50, 1000)$ :

$$m = \frac{1000 - 750}{50 - 25} = 10, y = 10x + b.$$



Using (25, 750):

$$750 = 10(25) + b \implies b = 500, y = 10x + 500.$$

Cost of 100 items:  $y = 10(100) + 500 = 1500$ .

#### Example A.15: Temperature Conversion

Freezing point of water:  $(C, F) = (0, 32)$ ; boiling point:  $(100, 212)$ .

**Solution:**

$$m = \frac{212 - 32}{100 - 0} = \frac{180}{100} = 1.8, F = 1.8C + 32.$$

For  $C = 30$ :  $F = 1.8(30) + 32 = 86$ .

## A.5 More Applications: Systems of Linear Equations

### Outcomes

- Solve linear systems in two variables.
- Find equilibrium points (supply = demand).
- Find break-even points (cost = revenue).

### Solving Systems of Equations

The intersection of two lines can be found by setting their equations equal or by using the elimination method.

#### Example A.16: Solving a System

Solve:

$$2x + y = 7, \quad 3x - y = 3.$$

**Solution:** Add them:

$$2x + y = 7, \quad 3x - y = 3 \implies 5x = 10 \implies x = 2.$$

Substitute  $x = 2$  into  $2x + y = 7 \implies 4 + y = 7 \implies y = 3$ .

**Solution:** (2, 3).

## Supply, Demand, and Equilibrium

The *equilibrium price* occurs where supply equals demand.

### Example A.17: Equilibrium Point

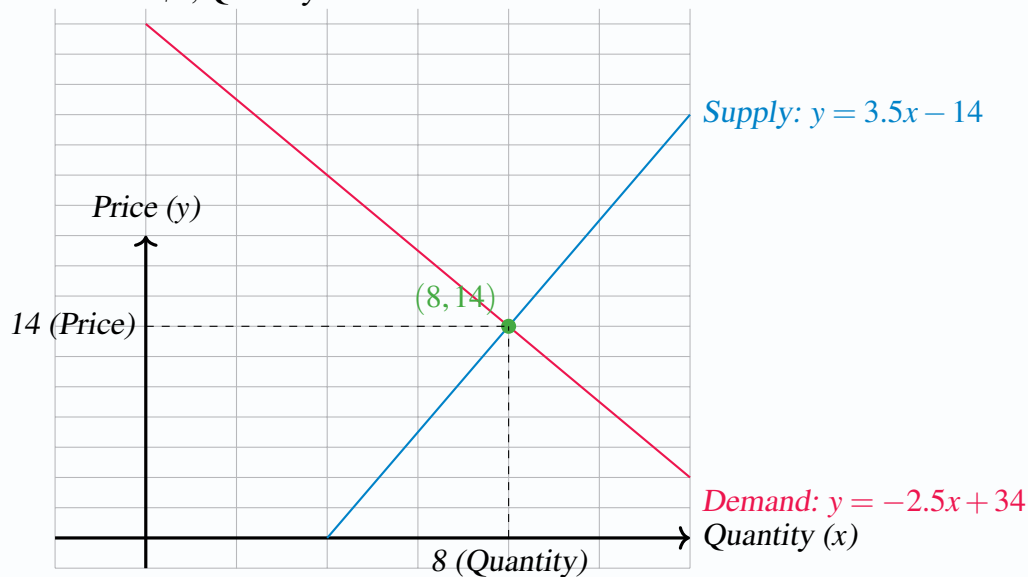
Supply:  $y = 3.5x - 14$ , Demand:  $y = -2.5x + 34$ .

**Solution:** Set equal:

$$3.5x - 14 = -2.5x + 34 \implies 6x = 48 \implies x = 8.$$

At  $x = 8$ :  $y = 3.5(8) - 14 = 14$  or  $y = -2.5(8) + 34 = 14$ .

Equilibrium: Price = \$8, Quantity = 14 items.



## Break-Even Point

For cost  $C$  and revenue  $R$ , the break-even point is where  $C = R$ .

### Example A.18: Break-Even Analysis

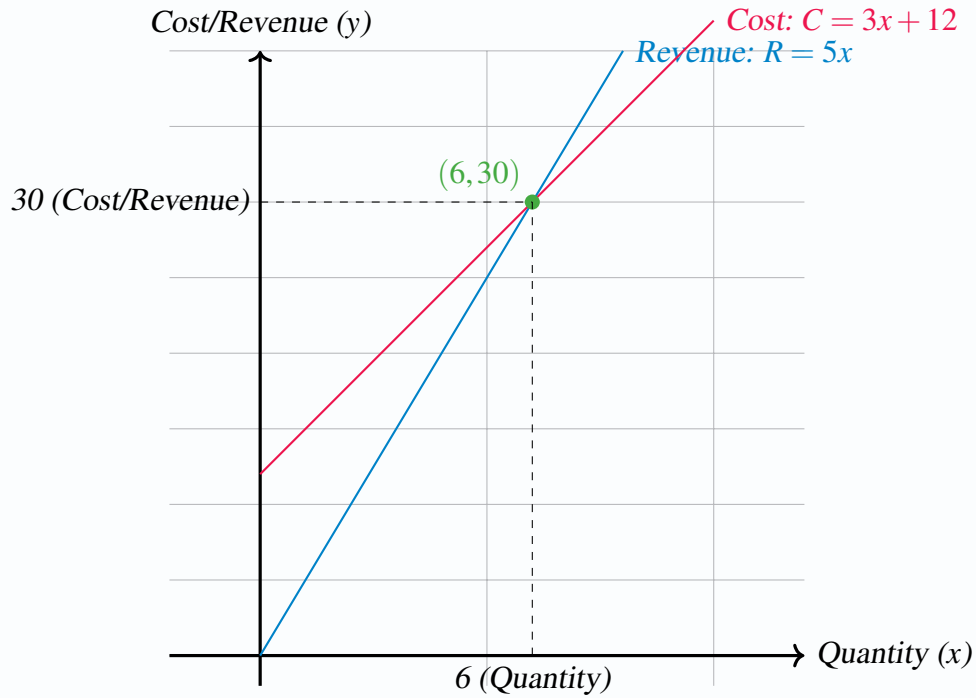
$R = 5x$ ,  $C = 3x + 12$ .

**Solution:**

$$5x = 3x + 12 \implies 2x = 12 \implies x = 6.$$

At  $x = 6$ ,  $R = C = 30$ .

Break-even at  $(6, 30)$ .



## Chapter Review

This chapter introduced the fundamental concepts of linear equations, their graphs, slopes, forms of equations, and their real-world applications. You can now model scenarios involving costs, pricing, and market equilibrium using linear models and solve these models for key insights.

